UNIT 03 · ACTIVITY 03

Case Study #4

A mobile app and a web app duplicate the same business rules and drift apart.



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DUPLICATED RULES. INCONSISTENT RESULTS.

SAME WORKSHOP · DIFFERENT TOTAL

WEB	MOBILE
149.00	148.50

A single rule has two implementations, so its behavior changes depending on the device.

THE USER GETS TWO ANSWERS



01 **DUPLICATED LOGIC**

Web and mobile enforce business rules independently.



02 **RULES DRIFT APART**

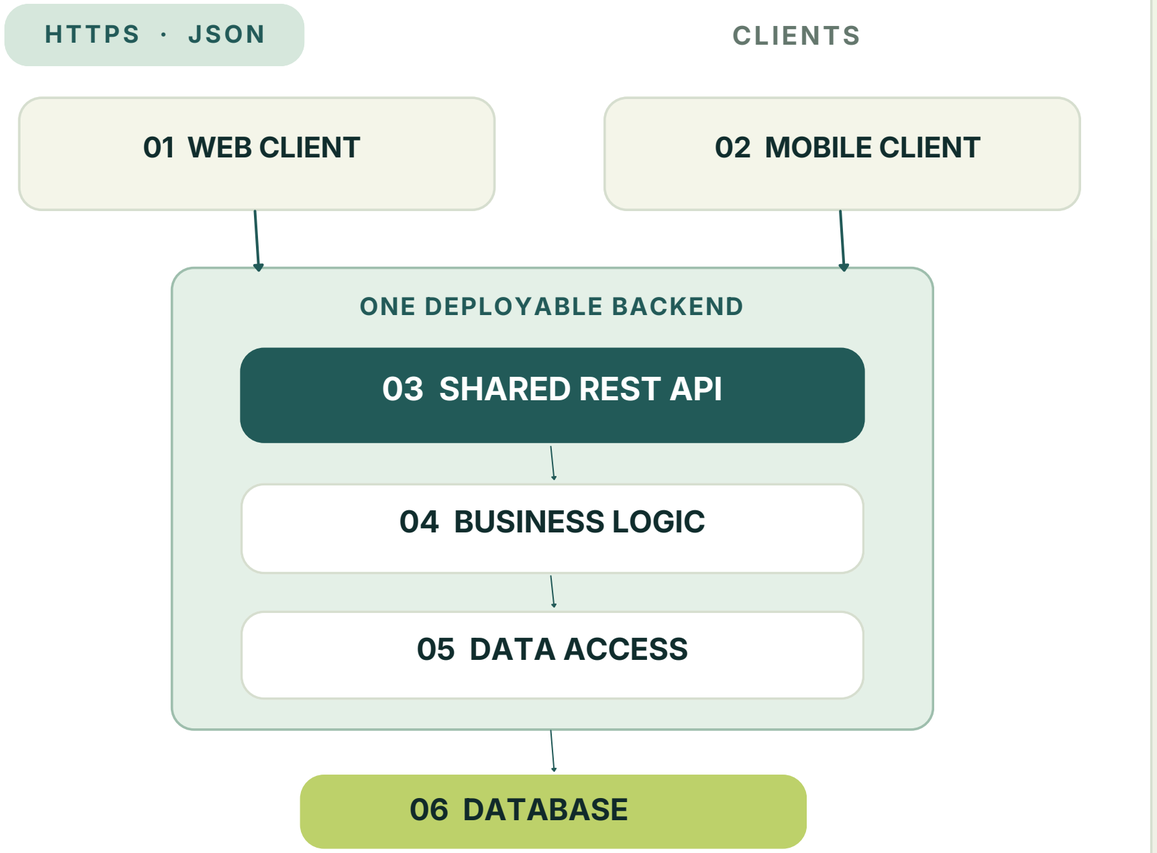
Small changes create different validations and totals.



03 **DOUBLE MAINTENANCE**

Teams update, test and release the same rule twice.

ONE REST API. SIX CLEAR ROLES.



01 — WEB CLIENT
Browser UI: collects input and shows results.

02 — MOBILE CLIENT
Phone UI: same contract and business outcomes.

03 — SHARED REST API
One entry point: auth, validation, JSON + status.

04 — BUSINESS LOGIC
One source of truth for calculations and rules.

05 — DATA ACCESS
Safe queries and transactions with persistence.

06 — DATABASE
Records and integrity constraints.

503

SERVICE UNAVAILABLE

No stale values presented as current.
No blind retries on writes.

```
HTTP/1.1 503 Service Unavailable
Retry-After: 30
Content-Type: application/problem+json

{ "type": "about:blank",
  "title": "Service unavailable",
  "status": 503,
  "detail": "Try again shortly.",
  "requestId": "8F3A" }
```

WHEN THE DATABASE GOES OFFLINE.

01 — DETECT

Connection timeouts, pool limits and health checks detect the failure.

02 — RETURN 503

The API returns a standard error plus Retry-After; no stack traces.

03 — INFORM CLIENTS

Show a clear message and label cached results instead of implying they are fresh.

04 — RECOVER SAFELY

Use request IDs, monitoring and a circuit breaker; never retry writes blindly.

NOT ANOTHER JOB BOARD.
A PATH TO THE JOB YOU WANT.

Students can find vacancies. What they need is clarity on the skills they lack — and what to do next.



01

MY PROFILE

WHAT YOU KNOW

Skills, experience and projects already in your portfolio.



02

JOB ANALYSIS

WHAT JOBS REQUIRE

Compare vacancy requirements with your actual skills.



03

MY LEARNING PATH

WHAT TO BUILD NEXT

Personalized learning goals and practical project evidence.

ENPOINTS

SKILL MATCHING MICROSERVICE

01 COMPLETE JOB ANALYSIS

400 · 401 · 413 · 500

POST

/api/v1/analyses

IN jobText · skills[] · technologyCatalog[]

OUT Matches · gaps · compatibility %

02 COMPARE SKILLS

400 · 401 · 500

POST

/api/v1/matches

IN skills[] · requirements[]

OUT Matches · gaps · compatibility %

03 EXTRACT TECHNOLOGIES

400 · 401 · 413 · 500

POST

/api/v1/extractions

IN jobText · technologyCatalog[]

OUT Technologies with catalog ID and mentions

Thank You

